

BHARAT BYTES

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UNDERSTANDING THE PROBLEM SPACE

Problem

- Identify high-risk customers
- Wrong predictions lead to financial loss or unfair pricing
- Dataset is imbalanced and anonymized

Goal

- Predict probability (0–1) of insurance claim
- Focus on ranking quality → Normalized Gini
- Generate predictions on test data only

Training data → 50 features + target
Test data → 50 features (no target)

Binary
Features

Categorical
Features

Numeric
Features

WHY CATBOOST?



CatBoost

- Ordered boosting → avoids leakage in rare events
- Native categorical handling → no unsafe encoding
- Stable under imbalance → preserves ranking

DATA PREPARATION & VALIDATION STRATEGY

Data Preprocessing

Separated target and identifiers

Removed ID column from training inputs

Explicit handling of categorical features

Feature Type Handling

Binary features kept as-is

Numerical features used directly

Categorical features:

- Converted to string
- Passed via CatBoost `cat_features`

Data Splitting Strategy

60% Training

20% Validation

20% Holdout

Stratified split to preserve class imbalance

Imbalance Handling

Claim events are rare

Ranking-based metric (Gini)

Oversampling avoided due to ranking and probability distortion

Focused on probability ranking

MODELING APPROACH OVERVIEW

Model Configuration

1. Tree depth = 8
2. Learning rate = 0.02
3. Bernoulli bootstrap
4. Early stopping (150 rounds)
5. 3000 max iterations
6. AUC as validation metric

Training Strategy

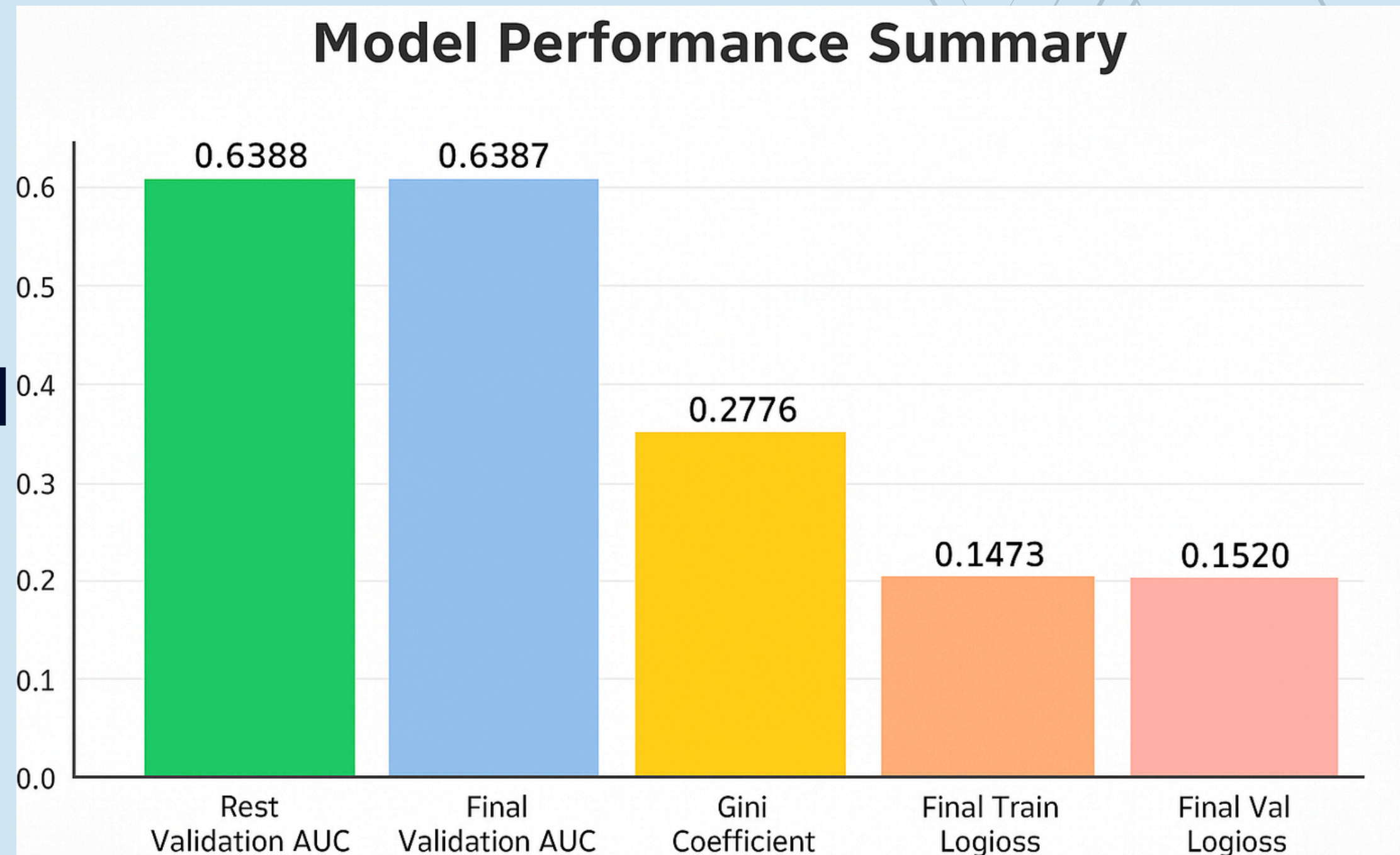
1. Train on train set, validate on holdout
2. Best iteration selected via early stopping
3. Final model retrained using optimal settings

Evaluation Metric

1. Normalized Gini coefficient
2. Measures ranking quality
3. Industry standard for insurance
4. Effective for imbalanced data

MODEL PERFORMANCE

- Validation AUC: 0.6388
- Normalized Gini: 0.2776
- Stable validation
- No severe overfitting observed
- Minimal gap between train and validation
- Industry benchmark Normalized Gini:



HOW ZERVE AI ENABLED OUR WORKFLOW

- Rapid experimentation without environment setup
- Entire pipeline as sequential DAG.
- Seamless execution of ML pipelines
- Easy iteration on preprocessing and models
- Centralized canvas for code and results

BUSINESS IMPACT & FUTURE ROADMAP

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Business Value

- Improved risk-based pricing
- Early identification of high-risk / fraudulent claims
- More efficient claim management
- Scalable and reusable ML pipeline

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Future Improvements

- Model ensembling for higher Gini
- Probability calibration for better risk interpretation
- Feature interaction and explainability analysis

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CONCLUSION

**ROBUST, LEAKAGE-SAFE, RANKING-OPTIMIZED ML PIPELINE
ALIGNED WITH REAL INSURANCE DECISION-MAKING**

That's all for our presentation.

THANKS

Thank you for your time and willingness to listen
to our explanation and narration.

